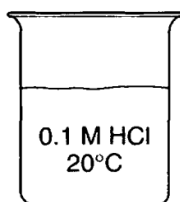


Unit 8 - Nuclear & Kinetics/Equilibrium Quiz [RETAKE]

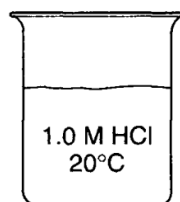
1. Which change would most likely decrease the rate of a chemical reaction because particles collide less often and with less energy?
 - a. Decreasing the surface area of the reactants
 - b. Lowering the concentration of reactants
 - c. decreasing the temperature of the system
 - d. Removing a catalyst
2. A reaction reaches dynamic equilibrium when
 - a. the concentrations of products and reactants become equal
 - b. the forward and reverse reactions stop
 - c. the rate of the forward reaction equals the rate of the reverse reaction
 - d. all reactants have been converted to products
3. Which statement best explains why powdered calcium carbonate reacts faster than a marble chip of the same mass?
 - a. Powdered calcium carbonate has greater surface area exposed for collisions
 - b. Powdered calcium carbonate has less mass
 - c. Powdered calcium carbonate lowers activation energy
 - d. Powdered calcium carbonate increases pressure
4. According to collision theory, a chemical reaction can occur only if reacting particles
 - a. collide with sufficient energy and proper orientation
 - b. collide at low temperatures only
 - c. contain equal amounts of reactants and products
 - d. remain in constant motion without colliding
5. Which change would most likely decrease the rate of a chemical reaction?
 - a. Increasing pressure in a gas reaction
 - b. Increasing temperature
 - c. Adding a catalyst
 - d. Decreasing the concentration of reactants
6. A reversible reaction is best defined as a reaction in which
 - a. products can reform reactants
 - b. products are permanently removed
 - c. only one direction of reaction is possible
 - d. energy is neither absorbed nor released
7. Which statement best describes activation energy?
 - a. The total energy stored in reactants
 - b. The minimum energy required for effective collisions
 - c. The energy released after products form
 - d. The energy needed to stop a reaction
8. When a catalyst is added to a reaction system at equilibrium, the catalyst
 - a. increases the activation energy of the forward reaction
 - b. decreases the activation energy of both the forward and reverse reactions
 - c. changes the equilibrium concentrations of products and reactants
 - d. stops the reverse reaction completely
9. In each of the four beakers shown below, a 2.0-centimeter strip of magnesium ribbon reacts with 100mL of HCl(aq) under the conditions shown:

In which beaker will the reaction occur at the slowest rate?

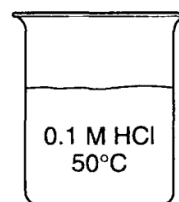
- a. A c. C
- b. B d. D



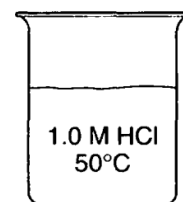
Beaker A



Beaker B



Beaker C



Beaker D

10. Which statement correctly describes an exothermic reaction?

- a. Heat is absorbed, causing the surroundings to cool
- b. Heat is released, causing the surroundings to decrease in Kinetic Energy
- c. Activation energy becomes negative
- d. Products contain less energy than reactants

11. Given the balanced equation representing a reaction: $\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$, how does the reaction rate differ when using strips of iron vs. using a solid piece of iron of the same mass? [1]

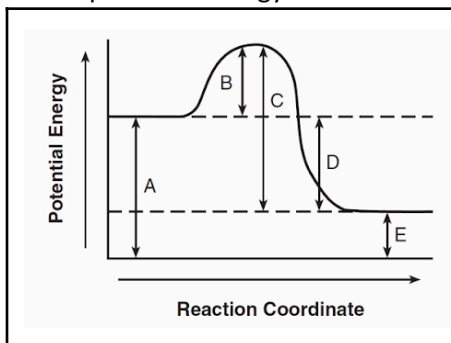
12. A student performs the same reaction twice, once at 20°C and once at 50°C. Explain why the reaction at 50°C occurs faster. Your answer must include the words "particles" and "energy." [1]

13. At room temperature, a reaction occurs when $\text{KIO}_3\text{(aq)}$ is mixed with $\text{NaHSO}_3\text{(aq)}$ that contains a small amount of starch. The colorless reaction mixture turns dark blue after a period of time that depends on the concentration of the reactants. Identify one factor, other than the surface area of the reactants, that would affect the rate of this reaction, and explain why. [1]

14. What are the three main components of collision theory? [1]

- a. _____ b. _____ c. _____

15. Which potential energy interval in the diagram represents the activation energy of the reverse reaction? [1]



Write your answer here:

16. Fill in the blank in each of the following decay reactions with the correct decay particle or decayed nucleus that will balance the reaction. [2]



17. Which statement best describes what happens in a fusion reaction? [1]

- (A) Energy is absorbed and more stable elements are formed.
- (B) Light nuclei form into heavier nuclei.
- (C) Heavy nuclei split into lighter nuclei.
- (D) Energy is released and less stable elements are formed.

